

Project Based Learning

On

RESIDENTIAL BUILDING AND DRAFTING USING AUTOCAD

By

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ABSTRACT

A residential building is a structure designed to provide safe, comfortable, and functional living spaces for occupants. Proper planning and drafting are essential for achieving efficient space utilization, structural safety, ventilation, lighting, and easy construction.

AutoCAD is widely used in civil engineering and architectural drafting to prepare accurate 2D drawings such as floor plans, elevations, sections, and detailed layouts. This project presents the design and drafting of a residential building using AutoCAD with proper dimensions, room arrangements, doors, windows, staircase, sanitary facilities, and other necessary building components.

The project focuses on developing a clear and accurate residential building plan following standard drafting practices. The drawing includes the floor plan, dimensions, room labels, wall thicknesses, openings, staircase details, and other architectural elements. Accurate dimensioning, labeling, and drafting help improve communication between designers, engineers, and construction workers.

INTRODUCTION

Residential buildings are one of the most common types of structures constructed to provide comfortable and safe living spaces for individuals and families. Proper planning of a residential building is important to ensure efficient utilization of available space, adequate ventilation and lighting, convenient movement, and functional arrangement of different rooms.

Building planning involves the preparation of floor plans, elevations, sections, and other necessary drawings. A properly prepared drawing provides clear information about the dimensions, location, and arrangement of various building components.

AutoCAD is an important computer-aided drafting software used in civil engineering and architectural applications. It helps in preparing accurate and professional drawings with precise dimensions and clear representation of building elements.

This project aims to design and draft a residential building using AutoCAD. The project includes the preparation of a residential floor plan with appropriate

room arrangements, walls, doors, windows, staircase, dimensions, and other architectural features.

The use of AutoCAD makes the drafting process faster and more accurate compared with manual drafting. It also allows easy modification of dimensions and design elements whenever required.

DRAWING

RESIDENTIAL BUILDING PLAN

BUILDING DETAILS:-

BUILDING – PLAN DETAILS

Description	**Size / Details**
Plot Size	32'-0" × 40'-0"
Plot Area	1280 sq.ft
Built-up Area	1050 sq.ft
Parking Area	225 sq.ft
Bedroom 1	13’-0” × 11’-0”
Bedroom 2	12’-0” × 11’-0”
Hall	15’-6” × 12’-0”
Dining Area	12’-0” × 10’-0”
Kitchen	9’-0” × 9’-0”
Utility	4’-6” × 9’-0”
Toilet 1	7’-0” × 4’-6”
Toilet 2	6’-6” × 4’-6”
Parking	15’-0” × 14’-6”
Main Gate	10’-0”

Description	Size / Details
Building Type	Residential building
Drawing Type	Ground Floor Plan
Units	Feet and Inches
Main Door	4’-0” × 7’-0”
Door 1	3’-6” × 7’-0”
Door 2	2’-6” × 7’-0”
Door 2	2’-6” × 7’-0”
Door 3	2’-0” × 7’-0”
Window 1	5’-0” × 4’-0”
Window 2	4’-0” × 4’-0”
Ventilators	2’-0” × 2’-0”
Staircase Window	3’-0” × 3’-0”

DIMENSION CALCULATIONS :-

Plot Length
40 × 304.8 = **12192 mm**

Plot Width
32 × 304.8 = **9753.6 mm**

Bedroom 1
13 × 304.8 = **3962.4 mm**
11 × 304.8 = **3352.8 mm**

Bedroom 2
12 × 304.8 = **3657.6 mm**
11 × 304.8 = **3352.8 mm**

Hall
15.5 × 304.8 = **4724.4 mm**
12 × 304.8 = **3657.6 mm**

Dining Area
12 × 304.8 = **3657.6 mm**
10 × 304.8 = **3048 mm**

Kitchen
9 × 304.8 = **2743.2 mm**

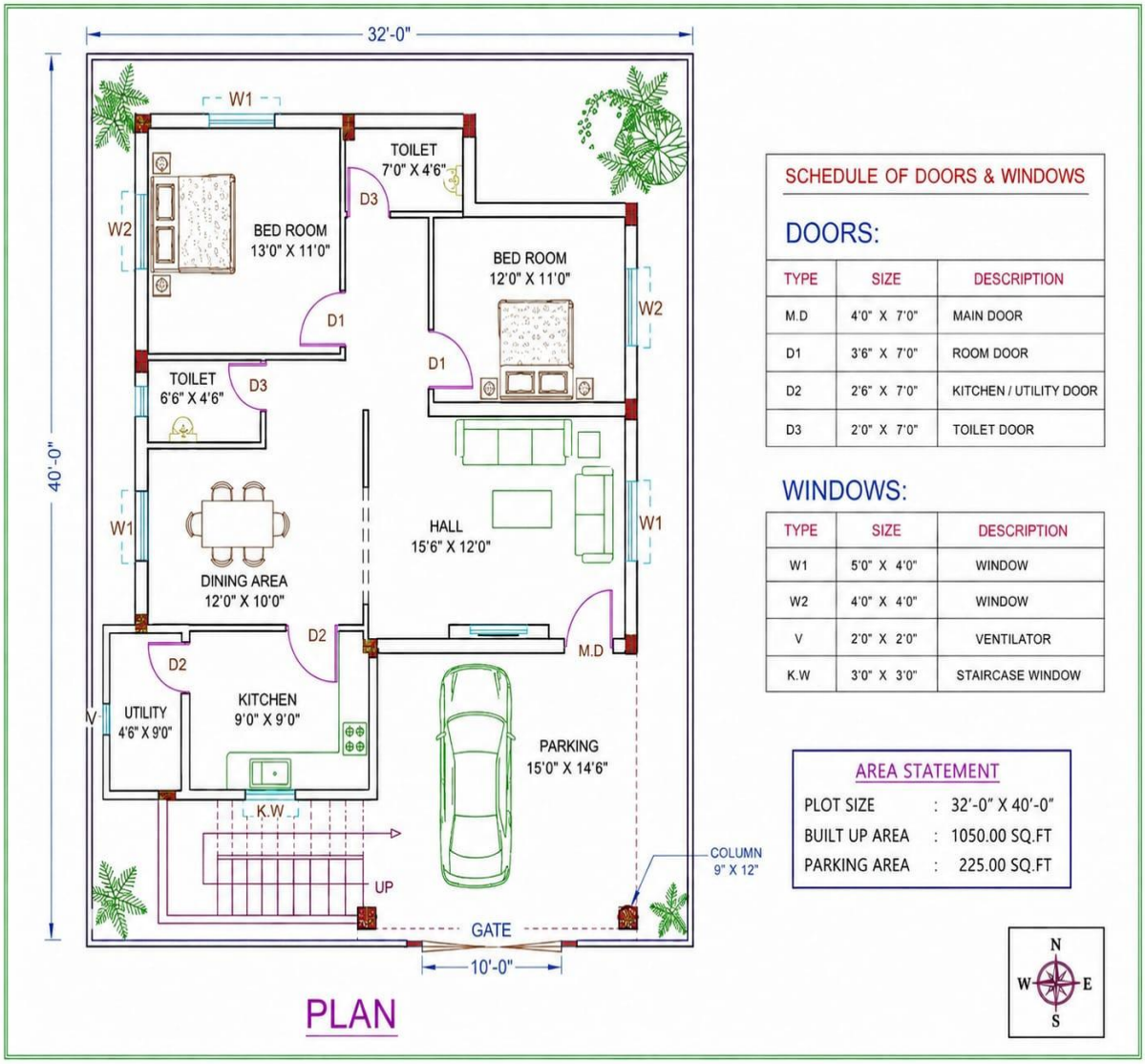
Utility
4.5 × 304.8 = **1371.6 mm**
9 × 304.8 = **2743.2 mm**

Toilet 1
7 × 304.8 = **2133.6 mm**
4.5 × 304.8 = **1371.6 mm**

Toilet 2
6.5 × 304.8 = **1981.2 mm**
4.5 × 304.8 = **1371.6 mm**

Main Gate
10 × 304.8 = **3048 mm**

RESIDENTIAL BUILDING DESIGN



ELEVATION INCLUDES

- Main entrance door
- Windows
- Ventilators
- External walls
- Sunshades
- Roof slab
- Parapet wall
- Staircase
- Columns
- Plinth level
- Ground level
- Building height
- Architectural features
- Required dimensions

AUTOCAD DRAWING TOOLS USED

Drawing Tools

- Line
- Polyline
- Rectangle
- Circle
- Arc
- Hatch
- Text
- MText
- Dimension
- Leader

Modify Tools

- Move
- Copy
- Offset
- Mirror
- Trim
- Extend
- Rotate
- Scale
- Stretch
- Array
- Fillet
- Chamfer
- Explode

TECHNOLOGY USED

Software

- AutoCAD 2024
- 2D Computer-Aided Design (CAD)
- Windows 10 / Windows 11

Drafting Technology

- Computer-aided architectural drafting
- 2D floor plan preparation
- Digital dimensioning and annotation
- Layer-based drafting
- Digital modification and editing

DESIGN STANDARDS

- National Building Code of India (NBC) – Guidelines for building planning and construction.
- IS 456:2000 – Plain and Reinforced Concrete.
- IS 875 – Design Loads for Buildings and Structures.
- IS 800 – General Construction in Steel.
- Standard architectural drafting and dimensioning practices.

APPLICATIONS

- Residential building planning
- Individual houses
- Apartments
- Housing layouts
- Building floor plans
- Building elevations and sections
- Architectural drafting
- Construction planning
- Building modification and renovation

ADVANTAGES

- Accurate and precise building drawings
- Easy modification of the design
- Saves drafting time
- Proper dimensioning and annotation
- Clear representation of building components
- Better utilization of available space
- Easy preparation of floor plans and elevations
- Reduces manual drafting errors
- Professional-quality drawings
- Useful for construction and estimation

CONCLUSION

The project demonstrates the design and drafting of a residential building using AutoCAD. The residential floor plan was prepared with proper dimensions and arrangements of bedrooms, hall, dining area, kitchen, utility, toilets, staircase, parking, doors, windows and other building components.

The use of AutoCAD helped in preparing an accurate, clear and professional 2D building drawing. Proper dimensioning and detailing make the drawing easier to understand and useful for construction planning. The project also provided practical knowledge of computer-aided drafting and residential building planning.